1.2. Well – formed Formulas

A ***well-formed formula*** (***wff***) is defined recursively as follows:

(i) *A proposition variable alone is a wff*
(ii) *If a is a wff then $\sim a$ is a wff*
(iii) *If a and b are wffs, then $(a \wedge b)$, $(a \vee b)$, $(a \longrightarrow b)$ and $(a \leftrightarrow b)$ are wffs.*
(iv) *A string of symbols containing the proposition variables, connectives and parentheses is a wff if and only if it can be obtained by finitely many applications of the above rules (i), (ii) and (iii).*

By the above definitions the following are wffs:

$$(p \wedge q)\ ,\big((p \wedge q) \longrightarrow q\big),\Big(\big((p \longrightarrow q) \wedge (q \longrightarrow r)\big) \leftrightarrow (p \longrightarrow r)\Big)$$

We use parentheses to avoid ambiguity. For the sake of convenience we shall omit the outer parentheses. Thus we write the above wffs respectively as

$$p \wedge q\ ,(p \wedge q) \longrightarrow q,\big((p \longrightarrow q) \wedge (q \longrightarrow r)\big) \leftrightarrow (p \longrightarrow r)$$

Since we deal with only wffs, we refer wffs as *formulas*.

The following are not formulas

$$p \vee q \wedge r\ ,\ \ (p \longrightarrow q) \longrightarrow (\wedge r),(p \wedge q) \longrightarrow q)$$

Note:

(i) In the construction of formulas, the parentheses will be used in the same sense in which they are used in elementary algebra or some times in a programming language.

(ii) The following is the hierarchy of operations and parentheses:

1. Connectives within parentheses; among parentheses innermost first
2. Negation $\sim$
3. $\wedge$ and $\vee$
4. $\longrightarrow$
5. $\leftrightarrow$

Truth tables of well-formed Formulas

If there are $\boldsymbol{n}$ distinct variables in a formula, then we need to consider $\mathbf{2^n}$ possible combinations of truth values in order to obtain the truth table of the formula.

Construction of truth tables

There are **two** methods of construction of truth tables.

The first columns of the table are for the proposition variables p, q, r, … and create enough number of rows ($i.e.$, 2^n rows if there are n *variables*) in the table for all possible combinations of T and F for the variables.

Method – 1

Devote a column for each elementary stage of the construction of the formula. The truth value at each step is determined from the previous stages by the definition of connectives. Finally the truth value of the formula is given in the last column.

Method – 2

Write the formula on the top row to the right of its variables. A column is drawn for each variable as well as for the connectives that appear in the formula. The truth values are entered step by step. The step numbers at the bottom of the table show the sequence followed in arriving the final step.

Example 1:

Construct the truth table for the formula

$$(p \vee q) \longrightarrow \big((r \vee p) \wedge (\sim r \vee q)\big)$$

Solution:

Method- 1

Let β be $(r \vee p) \wedge (\sim r \vee q)$. Then the given formula is $(p \vee q) \longrightarrow \beta$.

Truth table for $(p \vee q) \longrightarrow \big((r \vee p) \wedge (\sim r \vee q)\big)$								
p	q	r	$p \vee q$	$r \vee p$	$\sim r$	$\sim r \vee q$	β	$(p \vee q) \longrightarrow \beta$
T	T	T	T	T	F	T	T	T
T	T	F	T	T	T	T	T	T
T	F	T	T	T	F	F	F	F
T	F	F	T	T	T	T	T	T
F	T	T	T	T	F	T	T	T
F	T	F	T	F	T	T	F	F
F	F	T	F	T	F	F	F	T
F	F	F	F	F	T	T	F	T

Method- 2

$\boldsymbol{p}$	$\boldsymbol{q}$	$\boldsymbol{r}$	$(\boldsymbol{p}$	$\vee$	$\boldsymbol{q})$	$\longrightarrow$	$((\boldsymbol{r}$	$\vee$	$\boldsymbol{p})$	$\wedge$	$(\sim$	$\boldsymbol{r}$	$\vee$	$\boldsymbol{q}))$
T	T	T	T	T	T	T	T	T	T	T	F	T	T	T
T	T	F	T	T	T	T	F	T	T	T	T	F	T	T
T	F	T	T	T	F	F	T	T	T	F	F	T	F	F
T	F	F	T	T	F	T	F	T	T	T	T	F	T	F
F	T	T	F	T	T	T	T	T	F	T	F	T	T	T
F	T	F	F	T	T	F	F	F	F	F	T	F	T	T
F	F	T	F	F	F	T	T	T	F	F	F	T	F	F
F	F	F	F	F	F	T	F	F	F	F	T	F	T	F
1	**1**	**1**	**1**	**2**	**1**	**7**	**1**	**3**	**1**	**6**	**4**	**1**	**5**	**1**

Step 7 is the final step and it gives the truth values of the given formula

Tautology

A wff is said to be a ***tautology*** *(a* ***universally valid formula*** *or* ***logical truth****) if its truth value is T for all possible assignments of truth values to the proposition variables of the formula.*

Note:

(i) If p is any proposition then the formula $p \vee \sim p$ is a tautology
(ii) The conjunction of two tautologies is also a tautology

The proof of the above result (ii) is given below:

Let a and b be formulas which are tautologies. For all possible assignments of truth values to the variables of a and b, the truth values of both a and b will be T and hence the truth value of $a \wedge b$ will be T. Thus $a \wedge b$ is a tautology.

Contradiction

*A wff is said to be a **contradiction** (or **absurdity**) if its truth value is F for all possible assignments of truth values of the proposition variables of the formula.*

Note:

(i) If p is any proposition then the formula $p \wedge \sim p$ is a contradiction
(ii) The negation of a contradiction is a tautology

Contingency

*A wff is said to be **satisfiable** or **contingency** if it is neither a tautology nor a contradiction.*

Ex: If p and q are propositions then $p \wedge q$ and $p \vee q$ are satisfiable (contingency).

Example 2:

Show that the formula $\big(p \longrightarrow (q \longrightarrow r)\big) \longrightarrow \big((p \longrightarrow q) \longrightarrow (p \longrightarrow r)\big)$
is a tautology.

Solution:

Method-1

Let α be the formula $\big(p \longrightarrow (q \longrightarrow r)\big)$ and β be the formula $(p \longrightarrow q) \longrightarrow (p \longrightarrow r)$.
Then the given formula is $\alpha \longrightarrow \beta$.

Truth table for $\alpha \longrightarrow \beta$								
p	q	r	$q \longrightarrow r$	α	$p \longrightarrow q$	$p \longrightarrow r$	β	$\alpha \longrightarrow \beta$
T	T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	F	T
T	F	T	T	T	F	T	T	T
T	F	F	T	T	F	F	T	T
F	T	T	T	T	T	T	T	T
F	T	F	F	T	T	T	T	T
F	F	T	T	T	T	T	T	T
F	F	F	T	T	T	T	T	T

It is a tautology since its truth value is T for all possible assignments of truth values to the proposition variables of the formula.

Method- 2

$\boldsymbol{p}$	$\boldsymbol{q}$	$\boldsymbol{r}$	$(\boldsymbol{p}$	$\longrightarrow$	$(\boldsymbol{q}$	$\longrightarrow$	$\boldsymbol{r}))$	$\longrightarrow$	$((\boldsymbol{p}$	$\longrightarrow$	$\boldsymbol{q})$	$\longrightarrow$	$(\boldsymbol{p}$	$\longrightarrow$	$\boldsymbol{r}))$
T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T
T	T	F	T	F	T	F	F	T	T	T	T	F	T	F	F
T	F	T	T	T	F	T	T	T	T	F	F	T	T	T	T
T	F	F	T	T	F	T	T	T	T	F	F	T	T	F	F
F	T	T	F	T	T	T	T	T	F	T	T	T	F	T	T
F	T	F	F	T	T	F	T	T	F	T	T	T	F	T	F
F	F	T	F	T	F	T	T	T	F	T	F	T	F	T	T
F	F	F	F	T	F	T	T	T	F	T	F	T	F	T	F
1	**1**	**1**	**1**	**3**	**1**	**2**	**1**	**7**	**1**	**4**	**1**	**6**	**1**	**5**	**1**

Step 7 is the final step and it has truth values of T's only. Hence the given formula is a ***tautology***.

Example 3:

Show that the formula $\sim(p \vee q \vee \sim r) \wedge \big((r \longrightarrow p) \vee (r \longrightarrow q)\big)$ is a contradiction.

Solution:

We show this by constructing truth table.

p	q	r	$\sim$	$(p$	$\vee$	q	$\vee$	$\sim$	$r)$	$\wedge$	$((r$	$\longrightarrow$	$p)$	$\vee$	$(r$	$\longrightarrow$	$q))$
T	T	T	F	T	T	T	T	F	T	F	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T	T	F	F	F	T	T	T	F	T	T
T	F	T	F	T	T	F	T	F	T	F	T	T	T	T	T	F	F
T	F	F	F	T	T	F	T	T	F	F	F	T	T	T	F	T	F
F	T	T	F	F	T	T	T	F	T	F	T	F	F	T	T	T	T
F	T	F	F	F	T	T	T	T	F	F	F	T	F	T	F	T	T
F	F	T	T	F	F	F	F	F	T	F	T	F	F	F	T	F	F
F	F	F	F	F	F	F	T	T	F	F	F	T	F	T	F	T	F
1	**1**	**1**	**5**	**1**	**3**	**1**	**4**	**2**	**1**	**9**	**1**	**6**	**1**	**8**	**1**	**7**	**1**

Step 9 is the final step and it has truth values of F only. Hence the given formula is a contradiction.

Example 4:

Is the formula $(p \wedge q) \vee r \longrightarrow p \wedge (q \vee r)$ a tautology, contradiction or contingency?

Solution:

$\boldsymbol{p}$	$\boldsymbol{q}$	$\boldsymbol{r}$	$((\boldsymbol{p}$	$\wedge$	$\boldsymbol{q})$	$\vee$	$\boldsymbol{r})$	$\longrightarrow$	$(\boldsymbol{p}$	$\wedge$	$(\boldsymbol{q}$	$\vee$	$\boldsymbol{r}))$
T	T	T	T	T	T	T	T	T	T	T	T	T	T
T	T	F	T	T	T	T	F	T	T	T	T	T	F
T	F	T	T	F	F	T	T	T	T	T	F	T	T
T	F	F	T	F	F	F	F	T	T	F	F	F	F
F	T	T	F	F	T	T	T	F	F	F	T	T	T
F	T	F	F	F	T	F	F	T	F	F	T	T	F
F	F	T	F	F	F	T	T	F	F	F	F	T	T
F	F	F	F	F	F	F	F	T	F	F	F	F	F
1	**1**	**1**	**1**	**2**	**1**	**3**	**1**	**6**	**1**	**5**	**1**	**4**	**1**

Step 6 is the final step and it has truth values of T and F. Hence the given formula is neither a ***tautology*** nor a ***contradiction***. Therefore, it is a ***contingency***.

Substitution instance

Let a and b be formulas. We say that a is a ***substitution instance*** *of b if a can be obtained from b by substituting formulas for the variables of b, with the condition that the same formula is substituted for the same variable each time it occurs.*

Example: Let b: $p \longrightarrow (q \wedge p)$. Then

a: $(r \leftrightarrow s) \longrightarrow \big(q \wedge (r \leftrightarrow s)\big)$is a substitution instance of a , since $r \leftrightarrow s$ is substituted for p in b.Note that $(r \leftrightarrow s) \longrightarrow (q \wedge p)$ is not a substitution instance of b.

Note:

1. In constructing substitution instances of a formula, substitutions are made for the atomic formula and never for the molecular formula:

Ex: $p \longrightarrow q$ is not a substitution instance of $p \longrightarrow \sim r$, since it is r which must be substituted and not $\sim r$.

2. **Any substitution instance of a tautology is a tautology**

It is known that $p \vee \sim p$ is a tautology. If we substitute any formula for p, the resulting formula will be a tautology.

Example 5:

Determine the formulas which are substitution instances of other formulas in the following list and give the substitutions.

(a) $\big(p \longrightarrow (q \longrightarrow p)\big)$

(b) $\Big(\Big(\big((p \longrightarrow q) \wedge (r \longrightarrow s)\big) \wedge (p \vee r)\Big) \longrightarrow (q \vee s)\Big)$

(c) $\Big(q \longrightarrow \big((p \longrightarrow p) \longrightarrow q\big)\Big)$

(d) $\big(p \longrightarrow \big(p \longrightarrow (q \longrightarrow p)\big) \longrightarrow p\big)$

(e) $\left(\left(\left((r \longrightarrow s) \wedge (q \longrightarrow p)\right) \wedge (r \vee q)\right) \longrightarrow (s \vee p)\right)$

Solution:

(i) Substitute q for p and $(p \longrightarrow p)$ for q in (a), we get (c).Therefore ,(c) is the substitution instance of (a)

(ii) Substitute $\left(p \longrightarrow (q \longrightarrow p)\right)$for q in (a), we get (d). Therefore, (d) is the substitution instance of (a)

(iii) Substitute r, s, q and p for p, q, r and s respectively in (b) we get(e). Therefore, (e) is the substitution instance of $(\ b)$.

P1:

Construct the truth table for $(\boldsymbol{p} \vee \boldsymbol{q}) \vee \sim\boldsymbol{p}$.

Solution:

Truth table for $(p \vee q) \vee \sim p$				
p	q	$p \vee q$	$\sim p$	$(p \vee q) \vee \sim p$
T	T	T	F	T
T	F	T	F	T
F	T	T	T	T
F	F	F	T	T

P2:

Construct the truth table for $(\boldsymbol{p} \longrightarrow \boldsymbol{q}) \wedge (\boldsymbol{q} \longrightarrow \boldsymbol{p})$.

Solution:

Truth table for $(p \longrightarrow q) \wedge (q \longrightarrow p)$				
p	q	$p \longrightarrow q$	$q \longrightarrow p$	$(p \longrightarrow q) \wedge (q \longrightarrow p)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

P3:

Show that the formula $\sim(\boldsymbol{p} \wedge \boldsymbol{q}) \leftrightarrow (\sim\boldsymbol{p} \vee \sim\boldsymbol{q})$ is a tautology.

Solution:

We show this by constructing its truth table

Method- 1

Truth table for $\alpha : \sim(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$							
p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$(\sim p \vee \sim q)$	α
T	T	T	F	F	F	F	T
T	F	F	T	F	T	T	T
F	T	F	T	T	F	T	T
F	F	F	T	T	T	T	T

It is a tautology since its truth value is T for all possible assignments of truth values to the proposition variables of the formula.

Method- 2

$\boldsymbol{p}$	$\boldsymbol{q}$	$\sim$	$(\boldsymbol{p}$	$\wedge$	$\boldsymbol{q})$	$\leftrightarrow$	$(\sim$	$\boldsymbol{p}$	$\vee$	$\sim$	$\boldsymbol{q})$
T	T	F	T	T	T	T	F	T	F	F	T
T	F	T	T	F	F	T	F	T	T	T	F
F	T	T	F	F	T	T	T	F	T	F	T
F	F	T	F	F	F	T	T	F	T	T	F
1	**1**	**3**	**1**	**2**	**1**	**7**	**4**	**1**	**6**	**5**	**1**

Step 7 is the final step and it has truth values of T only. Hence the given formula is a ***tautology.***

$\therefore \sim(p \wedge q) \leftrightarrow (\sim p \vee \sim q)$ is a tautology

P4:

If p, q and r are proposition variables then show that $(p \longrightarrow q) \vee (\sim p \longrightarrow r)$ is a tautology.

Solution:

We show this by constructing truth table.

Method-1

Truth Table for $(p \longrightarrow q) \vee (\sim p \longrightarrow r)$							
p	q	r	$a: p \longrightarrow q$	$\sim p$	r	$b: \sim p \longrightarrow r$	$a \vee b$
T	T	T	T	F	T	T	T
T	T	F	T	F	F	T	T
T	F	T	F	F	T	T	T
T	F	F	F	F	F	T	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	T	T	T	T	T
F	F	F	T	T	F	F	T

It is a tautology since its truth value is T for all possible assignments of truth values to the proposition variables of the formula.

Method-2

p	q	r	$(p$	$\longrightarrow$	$q)$	$\vee$	$(\sim$	p	$\longrightarrow$	$r)$
T	T	T	T	T	T	T	F	T	T	T
T	T	F	T	T	T	T	F	T	T	F
T	F	T	T	F	F	T	F	T	T	T
T	F	F	T	F	F	T	F	T	T	F
F	T	T	F	T	T	T	T	F	T	T
F	T	F	F	T	T	T	T	F	F	F
F	F	T	F	T	F	T	T	F	T	T
F	F	F	F	T	F	T	T	F	F	F
1	**1**	**1**	**1**	**2**	**1**	**5**	**3**	**1**	**4**	**1**

Step 5 is the final step and it has truth values of T only. Hence the given formula is a ***tautology***.

P5:

Show that the formula $(\sim q \wedge p) \wedge (p \longrightarrow q)$ is a contradiction.

Solution:

p	q	$(\sim$	q	$\wedge$	$p)$	$\wedge$	$(p$	$\longrightarrow$	$q)$
T	T	F	T	F	T	F	T	T	T
T	F	T	F	T	T	F	T	F	F
F	T	F	T	F	F	F	F	T	T
F	F	T	F	F	F	F	F	T	F
1	**1**	**2**	**1**	**3**	**1**	**5**	**1**	**4**	**1**

Step 5 is the final step and it has truth values F only. Hence the given formula is a Contradiction

P6:

Show that the formula for $(p \wedge q) \wedge \sim(p \vee q)$ is a contradiction.

Solution:

p	q	$(p$	$\wedge$	$q)$	$\wedge$	$\sim$	$(p$	$\vee$	$q)$
T	T	T	T	T	F	F	T	T	T
T	F	T	F	F	F	F	T	T	F
F	T	F	F	T	F	F	F	T	T
F	F	F	F	F	F	F	F	T	F
1	**1**	**1**	**2**	**1**	**5**	**4**	**1**	**3**	**1**

Step 5 is the final step and it has truth values F only. Hence the given formula is a Contradiction

P7:

Show that the formula $(p \longrightarrow (q \longrightarrow r)) \longrightarrow ((p \longrightarrow q) \longrightarrow r)$ is a contingency.

Solution:

p	q	r	$(p$	$\longrightarrow$	$(q$	$\longrightarrow$	$r))$	$\longrightarrow$	$((p$	$\longrightarrow$	$q)$	$\longrightarrow$	$r)$
T	T	T	T	T	T	T	T	T	T	T	T	T	T
T	T	F	T	F	T	F	F	T	T	T	T	F	F
T	F	T	T	T	F	T	T	T	T	F	F	T	T
T	F	F	T	T	F	T	F	T	T	F	F	T	F
F	T	T	F	T	T	T	T	T	F	T	T	T	T
F	T	F	F	T	T	F	F	F	F	T	T	F	F
F	F	T	F	T	F	T	T	T	F	T	F	T	T
F	F	F	F	T	F	T	F	F	F	T	F	F	F
1	**1**	**1**	**1**	**3**	**1**	**2**	**1**	**6**	**1**	**4**	**1**	**5**	**1**

Step 6 is the final step and it has truth values T and F. Thus it is neither a Tautology nor a contradiction. Therefore it is a contingency.

P8:

Show that the formula $(p \longrightarrow (q \vee r)) \longrightarrow ((p \wedge q) \longrightarrow r)$ is a contingency.

Solution:

p	q	r	$(p$	$\longrightarrow$	$(q$	$\vee$	$r))$	$\longrightarrow$	$((p$	$\wedge$	$q)$	$\longrightarrow$	$r)$
T	T	T	T	T	T	T	T	T	T	T	T	T	T
T	T	F	T	T	T	T	F	F	T	T	T	F	F
T	F	T	T	T	F	T	T	T	T	F	F	T	T
T	F	F	T	F	F	F	F	T	T	F	F	T	F
F	T	T	F	T	T	T	T	T	F	F	T	T	T
F	T	F	F	T	T	T	F	T	F	F	T	T	F
F	F	T	F	T	F	T	T	T	F	F	F	T	T
F	F	F	F	T	F	F	F	T	F	F	F	T	F
1	**1**	**1**	**1**	**3**	**1**	**2**	**1**	**6**	**1**	**4**	**1**	**5**	**1**

Step 6 is the final step and it has truth values T and F. Thus it is neither a Tautology nor a contradiction. Therefore it is a contingency.

1.2. Well-Formed Formulas

Exercises:

Which of the following formulas is a tautology, contradiction, contingency?

a. $\sim\big((p \longrightarrow q) \longrightarrow ((r \vee p) \longrightarrow (r \vee q))\big)$

b. $(q \wedge (p \longrightarrow q)) \longrightarrow p$

c. $(p \vee q) \longrightarrow ((p \vee r) \vee (r \vee q))$

d. $\sim(p \vee (q \wedge r)) \leftrightarrow ((p \vee q) \wedge (p \vee r))$

e. $(p \vee (q \longrightarrow r)) \leftrightarrow ((p \vee \sim r) \longrightarrow q)$

f. $p \longrightarrow \Big(q \longrightarrow \big(r \longrightarrow (\sim p \longrightarrow (\sim q \longrightarrow \sim r))\big)\Big)$

g. $(p \longrightarrow \sim p) \longrightarrow \sim p$

h. $\big((p \wedge (p \longrightarrow \sim q)) \vee (q \longrightarrow \sim q)\big) \longrightarrow \sim q$

i. $(p \longrightarrow q) \longrightarrow ((p \vee r) \longrightarrow (q \vee r))$

j. $((p \vee q) \wedge \sim r) \longrightarrow \sim p \vee r$

k. $(\sim(p \vee q \vee \sim r)) \wedge ((r \longrightarrow p) \vee (r \longrightarrow q))$